Supplementary Material

(NAMAT: Node-wise Message-Aware Multi-Graph Attention for Cancer Driver Gene Prioritization)

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1 Supplementary Methods

1.1 Dataset construction and label definition

Twelve cancer-specific benchmark datasets were utilized, adhering to the experimental settings established by EMOGI[1] and MTGCN[2]. For each cancer type, positive samples were defined as known driver genes annotated in the Network of Cancer Genes (NCG)[3]. Negative samples were defined as passenger genes obtained from the MTGCN benchmark after excluding genes associated with established cancer-related resources, including NCG, COSMIC[4], OMIM[5], and KEGG cancer pathways[6]. This conservative filtering strategy minimizes the risk of misclassifying known cancer-associated genes as negatives.

A consistent set of 2,187 passenger genes was applied across all cancer types. The number of positive driver genes varied by cancer type, reflecting differences in available driver-gene annotations. Dataset statistics are provided in Supplementary Table S1.

Table S1. Dataset statistics across 12 cancer types. Positive samples correspond to driver genes, and negative samples correspond to passenger genes.

	BRCA	BLCA	LUAD	THCA	LIHC	LUSC	ESCA	PRAD	STAD	COAD	UCEC	CESC
Driver genes	202	95	179	64	82	26	71	50	73	141	68	17
Passenger genes	2187	2187	2187	2187	2187	2187	2187	2187	2187	2187	2187	2187

1.2 PPI network construction

We used the same six PPI networks as the EMOGI benchmark: CPDB[7], STRING-db[8], MultiNet[9], PCNet[10], IRefIndex, and IRefIndex_2015[11].

1.3 Implementation details and hyperparameters

NAMAT was implemented using PyTorch. The input feature encoder consisted of a three-layer multilayer perceptron with hidden dimensions of 64 and an output dimension of 64. The graph backbone incorporated two message-passing layers with hidden dimensions of 64 and 128. Dropout with a probability of 0.4 was applied to the input encoder, message-passing layers, and prediction head. Additionally, feature-wise dropout with a probability of 0.2 was applied to the input features during training. Prior to model input, node features were standardized using column-wise z-score normalization.

The model was optimized using the AdamW optimizer with weight decay. The base learning rate was set to 1×10^{-3}, employing a linear warm-up for 5 epochs followed by cosine decay. Weight decay was set to 4×10^{-4}. The training objective utilized weighted binary cross-entropy with logits, a label smoothing coefficient of $\varepsilon = 0.06$, and a positive-class weight $w = \min(N_{\text{neg}}/N_{\text{pos}}, 10)$ to address class imbalance.

To stabilize training, graph-attention weights were fixed to a uniform distribution over valid protein-protein interaction (PPI) networks during the first 30 epochs. Following this gate warm-up stage, PPI dropout was activated with a probability of 0.25 during attention computation. The attention temperature was annealed from 0.8 to 0.6 throughout training. Gradient clipping was applied with a maximum norm of 1.0. Early stopping was based on validation loss with a patience of 80 epochs, and the model parameters corresponding to the lowest validation loss were selected for final evaluation.

All experiments were conducted using 5-fold stratified cross-validation and 10 independent runs with different random seeds. Performance was reported as the mean and standard deviation of test area under the precision-recall curve (AUPRC) across all folds and runs.

1.4 Baseline comparison protocol

NAMAT was compared with six graph-based driver gene prioritization baselines: EMOGI[1], MTGCN[2], MRNGCN[12], DGHNN[13], MODIG[14], and EMGNN[15]. These methods represent complementary graph-learning strategies for cancer driver gene prioritization, including single-protein–protein interaction (PPI) graph convolution, multi-task graph learning, heterogeneous multi-network graph convolution, graph and hypergraph modeling, multi-dimensional biological network integration, and multilayer PPI graph learning.

To ensure a fair comparison, all methods were evaluated on the same 12 cancer-specific benchmark datasets, utilizing identical driver-gene labels, passenger-gene labels, gene-level data splits, and evaluation metrics, as supported by each method. The area under the precision-recall curve (AUPRC) served as the primary classification metric. For ranking-based evaluation, Hits@K was calculated from predicted gene scores using a consistent candidate-pool definition across all methods.

MRNGCN was included in the AUPRC comparison but excluded from the Hits@K analysis due to the unavailability of a runnable implementation for generating ranked candidate lists. Consequently, the Hits@K comparison comprises NAMAT, EMOGI, MTGCN, DGHNN, MODIG, and EMGNN.

1.5 Derivation of the –`MAA` ablation

Message-aware attention is a component of NAMAT that assigns node-wise reliability weights to different protein-protein interaction (PPI) networks. In the complete model, NAMAT does not assume equal informativeness across all PPI networks for every gene. For each node and each message-passing layer, the attention module utilizes graph-specific evidence, such as the propagated message from each PPI network and the graph-specific degree term, to determine the contribution of each network to the updated node representation.

The `−MAA` ablation eliminates the message-aware attention input while maintaining multi-graph message passing. This ablation is designed to determine whether NAMAT derives benefit from learning adaptive node-wise graph reliability weights, as opposed to merely utilizing multiple PPI networks. Consequently, `−MAA` evaluates the scenario in which all valid PPI networks contribute uniformly to each node.

In the full NAMAT model, the graph-specific evidence descriptor is defined as

$$\phi_{i,k}^{(t)} = \left[m_{i,k}^{(t)} \parallel \log(1+\deg_k(i)) \right], \tag{1}$$

and the attention input is

$$u_{i,k}^{(t)} = \left[h_i^{(t)} \parallel \phi_{i,k}^{(t)} \right]. \tag{2}$$

The attention score is then computed as

$$s_{i,k}^{(t)} = g^{(t)} \left(u_{i,k}^{(t)} \right). \tag{3}$$

Because $u_{i,k}^{(t)}$ contains graph-specific information, different PPI networks can receive different attention scores for the same node.

In the `−MAA` setting, the attention input is restricted to the current node state only:

$$u_{i,k}^{(t)} = h_i^{(t)}, \quad \forall k. \tag{4}$$

For a fixed node i and layer t, $h_i^{(t)}$ is identical across all valid graphs. Therefore, all valid graphs receive the same attention score:

$$s_{i,k}^{(t)} = g^{(t)} \left(h_i^{(t)} \right), \quad \forall k. \tag{5}$$

Applying the softmax over identical scores gives a uniform distribution over valid PPI networks:

$$\alpha_{i,k}^{(t)} = \frac{1}{K_i}, \tag{6}$$

where K_i denotes the number of valid PPI networks for node i. Thus, the aggregated message becomes

$$\tilde{m}_i^{(t)} = \frac{1}{K_i} \sum_{k=1}^{K_i} m_{i,k}^{(t)}. \tag{7}$$

Thus, `−MAA` corresponds to uniformly averaging graph-specific messages rather than learning node-wise graph reliability. Notably, this ablation does not eliminate graph message passing. Each PPI network continues to generate its own graph-specific message $m_{i,k}^{(t)}$. However, the model's capacity to assign distinct reliability weights to different PPI networks for a given node is removed. Consequently, the performance difference between NAMAT and `−MAA` reflects the impact of message-aware adaptive graph weighting.

1.6 Ablation experiment configuration

We conducted controlled ablation experiments to isolate the contribution of each major NAMAT component. The components evaluated were message-aware attention, the input MLP encoder, the residual update, and graph-specific message information.

The `-MAA` ablation removes message-aware adaptive graph weighting. In this configuration, graph-specific messages are still computed from all PPI networks, but the attention weights across valid networks are uniform rather than learned from message-specific evidence. This approach tests whether node-wise graph reliability estimation provides additional benefit beyond simple multi-network averaging.

The `-MLP` ablation removes the nonlinear input feature encoder. In this configuration, the model does not utilize the three-layer MLP to transform the original multi-omics feature vector prior to graph propagation. This experiment evaluates whether nonlinear feature transformation enhances driver gene prediction before message passing.

The `-Res` ablation removes the residual update from the message-passing layer. In this configuration, the next-layer representation is computed solely from the aggregated graph message, without incorporating a transformed version of the previous node representation. This experiment assesses whether preserving node-specific information across layers improves prediction.

The `-MSG` ablation removes graph-specific message information from the adaptive weighting mechanism. This configuration tests whether the attention mechanism benefits from directly observing the messages propagated by each PPI network. When message information is excluded, the model cannot estimate graph reliability based on the network-specific evidence received by each node.

All ablation variants were evaluated using the same datasets, train/validation/test splits, optimization protocol, and evaluation metrics as the full NAMAT model. This approach ensures that observed performance differences are attributable to the removed model component rather than variations in the experimental setting.

2 Supplementary Results

2.1 Detailed AUPRC Performance

Table S2. Detailed AUPRC comparison for 12 cancer types. MTG, DGH, and MRNG denote MTGCN, DGHNN, and MRNGCN, respectively. Results are reported as mean AUPRC $\pm$ standard deviation over repeated runs, rounded to three decimals. The top result for each cancer type is shown in **bold**. A superscript * indicates that NAMAT significantly outperformed EMGNN according to the Wilcoxon signed-rank test ($p < 0.05$).

Cancer	NAMAT	EMOGI	MTG	DGH	MRNG	MODIG	EMGNN
BRCA	**0.802**$\pm$0.006*	0.648$\pm$0.005	0.658$\pm$0.004	0.683$\pm$0.006	0.692$\pm$0.004	0.586$\pm$0.005	0.766$\pm$0.008
BLCA	**0.789**$\pm$0.005*	0.549$\pm$0.010	0.657$\pm$0.008	0.694$\pm$0.006	0.725$\pm$0.005	0.681$\pm$0.025	0.752$\pm$0.018
LUAD	**0.744**$\pm$0.008*	0.559$\pm$0.011	0.628$\pm$0.010	0.641$\pm$0.009	0.629$\pm$0.012	0.603$\pm$0.003	0.710$\pm$0.009
LIHC	**0.595**$\pm$0.019*	0.385$\pm$0.019	0.465$\pm$0.020	0.481$\pm$0.019	0.547$\pm$0.015	0.253$\pm$0.025	0.546$\pm$0.021
THCA	**0.426**$\pm$0.026*	0.259$\pm$0.008	0.291$\pm$0.011	0.315$\pm$0.023	0.308$\pm$0.010	0.028$\pm$0.001	0.362$\pm$0.032
LUSC	**0.616**$\pm$0.053*	0.222$\pm$0.027	0.310$\pm$0.041	0.405$\pm$0.054	0.348$\pm$0.032	0.071$\pm$0.014	0.524$\pm$0.048
ESCA	**0.611**$\pm$0.015*	0.417$\pm$0.030	0.472$\pm$0.028	0.476$\pm$0.015	0.537$\pm$0.012	0.264$\pm$0.019	0.568$\pm$0.023
PRAD	**0.827**$\pm$0.009*	0.523$\pm$0.022	0.614$\pm$0.019	0.706$\pm$0.013	0.669$\pm$0.016	0.383$\pm$0.097	0.783$\pm$0.035
STAD	**0.779**$\pm$0.009*	0.460$\pm$0.015	0.593$\pm$0.012	0.645$\pm$0.008	0.646$\pm$0.011	0.552$\pm$0.051	0.743$\pm$0.014
COAD	**0.610**$\pm$0.008*	0.346$\pm$0.007	0.382$\pm$0.009	0.447$\pm$0.069	0.495$\pm$0.007	0.443$\pm$0.066	0.587$\pm$0.006
UCEC	**0.743**$\pm$0.014	0.400$\pm$0.014	0.495$\pm$0.015	0.534$\pm$0.025	0.570$\pm$0.015	0.389$\pm$0.080	0.726$\pm$0.025
CESC	**0.655**$\pm$0.030	0.555$\pm$0.052	0.597$\pm$0.053	0.564$\pm$0.041	0.534$\pm$0.069	0.059$\pm$0.046	0.640$\pm$0.042

Supplementary Table S2 presents detailed AUPRC results for 12 cancer types. Results are summarized as mean AUPRC $\pm$ standard deviation from repeated runs. NAMAT achieved the highest mean AUPRC for all cancer types. As EMGNN was the strongest baseline, a Wilcoxon signed-rank test was conducted to compare NAMAT and EMGNN for each cancer type. A superscript * indicates that NAMAT significantly surpassed EMGNN at $p < 0.05$. NAMAT demonstrated a statistically significant improvement over EMGNN in 10 of 12 cancer types. For UCEC and CESC, NAMAT yielded a higher mean AUPRC, although this difference was not statistically significant.

2.2 Comprehensive Hits@K performance analysis

Supplementary Table S3 presents the complete Hits@K results for K=1,3,5,10,20,50 across 12 cancer types. Lower values of K reflect stringent top-ranked driver gene prioritization, while higher values of K indicate broader identification of known driver genes within larger candidate sets. NAMAT demonstrated the highest average Hits@K across all evaluated thresholds. MRNGCN was excluded due to the lack of a runnable implementation for generating ranked candidate lists.

Table S3. Comparison of Hits@K across 12 cancer types for K=1,3,5,10,20,50. Higher values correspond to greater recovery of known driver genes within the top-K ranked candidates. The best available results are highlighted in **bold**, and unavailable results are indicated by "–".

K	Method	BRCA	BLCA	LUAD	LIHC	THCA	LUSC	ESCA	PRAD	STAD	COAD	UCEC	CESC	Avg
1	NAMAT	**0.46**	**0.36**	0.28	**0.26**	**0.40**	**0.20**	**0.86**	**0.28**	0.28	**0.94**	0.44	**0.20**	**0.41**
	EMOGI	0.26	0.16	0.06	0.08	0.00	0.10	0.04	0.12	0.06	0.16	0.06	0.00	0.09
	MTGCN	0.20	0.16	0.20	0.08	0.04	0.02	0.02	0.04	0.04	0.02	0.06	0.00	0.07
	DGHNN	0.20	0.00	0.20	0.00	0.00	0.00	0.20	0.20	0.20	0.00	0.40	0.00	0.12
	MODIG	0.32	0.30	0.28	0.12	0.00	0.00	0.40	0.12	0.14	0.40	0.38	0.00	0.21
	EMGNN	0.40	0.26	**0.34**	0.20	0.36	0.18	0.58	0.26	**0.34**	0.86	**0.52**	0.18	0.37
3	NAMAT	**1.08**	**1.74**	**0.72**	**0.80**	**0.68**	0.36	**1.60**	0.82	**1.26**	**2.14**	**1.22**	**0.26**	**1.06**
	EMOGI	0.60	0.34	0.34	0.20	0.00	0.10	0.14	0.22	0.30	0.34	0.38	0.00	0.25
	MTGCN	0.38	0.24	0.26	0.16	0.08	0.02	0.18	0.08	0.22	0.18	0.12	0.06	0.17
	DGHNN	0.60	0.40	0.20	0.00	0.00	**0.40**	0.20	0.40	0.20	0.00	0.40	0.00	0.23
	MODIG	0.92	1.00	0.56	0.20	0.00	0.00	0.46	0.26	0.40	1.12	0.60	0.00	0.46
	EMGNN	**1.08**	1.40	0.66	0.78	0.66	0.30	1.18	**0.96**	1.14	1.80	1.06	0.20	0.94
5	NAMAT	1.86	**2.70**	**1.00**	**0.96**	**0.80**	0.44	**2.08**	1.06	**1.46**	**2.58**	**1.68**	0.68	**1.44**
	EMOGI	0.96	0.42	0.38	0.30	0.00	0.12	0.14	0.32	0.64	0.42	0.48	0.00	0.35
	MTGCN	0.80	0.36	0.50	0.22	0.08	0.04	0.26	0.18	0.50	0.40	0.26	0.14	0.31
	DGHNN	1.00	0.80	0.20	0.00	0.20	**0.60**	0.40	0.80	0.40	0.20	0.80	0.00	0.45
	MODIG	0.92	1.20	0.68	0.26	0.00	0.00	0.54	0.52	0.72	1.46	0.64	0.00	0.58
	EMGNN	**1.92**	2.28	0.96	0.92	0.68	0.36	1.30	**1.10**	1.36	2.14	1.60	**0.82**	1.29
10	NAMAT	**3.66**	**4.44**	**1.48**	1.18	**1.04**	0.56	**2.72**	**2.12**	**2.14**	**3.22**	**2.76**	**1.10**	**2.20**
	EMOGI	1.44	0.56	0.78	0.36	0.20	0.14	0.32	0.44	0.76	0.56	0.74	0.02	0.53
	MTGCN	1.66	0.56	0.84	0.24	0.16	0.08	0.30	0.36	0.64	0.58	0.50	0.24	0.51
	DGHNN	1.60	1.40	0.40	0.20	0.20	**0.60**	0.60	1.80	1.00	0.20	1.40	0.40	0.82
	MODIG	1.52	1.80	0.96	0.46	0.00	0.00	0.72	0.92	1.26	1.66	0.88	0.00	0.85
	EMGNN	3.44	3.98	1.42	**1.28**	0.84	0.50	1.78	1.48	1.90	2.82	2.40	1.06	1.91
20	NAMAT	**4.52**	**5.90**	**2.28**	1.72	**1.26**	0.70	**3.36**	**2.70**	**3.24**	**4.12**	**3.38**	**1.20**	**2.87**
	EMOGI	1.82	0.62	1.04	0.46	0.26	0.16	0.40	0.52	0.86	0.80	1.04	0.04	0.67
	MTGCN	2.20	0.72	1.34	0.24	0.22	0.12	0.56	0.82	0.76	0.72	0.76	0.34	0.73
	DGHNN	2.60	2.00	1.40	0.40	0.20	0.60	0.80	2.60	1.80	0.20	1.80	0.40	1.23
	MODIG	2.40	3.40	1.46	0.60	0.00	0.00	0.86	1.00	1.86	2.12	1.02	0.08	1.23
	EMGNN	4.40	5.10	1.98	**1.82**	1.22	**0.72**	2.62	2.58	2.90	3.84	2.94	1.18	2.61
50	NAMAT	**8.10**	**7.30**	**4.78**	**3.52**	**1.60**	**1.00**	**3.94**	**4.26**	**4.82**	**5.38**	**4.76**	1.34	**4.23**
	EMOGI	3.06	1.34	2.04	0.76	0.44	0.20	0.92	1.08	1.36	1.54	1.40	0.34	1.21
	MTGCN	3.20	1.54	2.22	0.40	0.26	0.22	0.96	1.26	1.46	1.14	1.08	0.48	1.19
	DGHNN	4.40	3.20	3.60	1.40	0.44	0.60	1.00	3.80	3.22	0.60	2.80	0.80	2.15
	MODIG	3.96	5.80	3.46	1.00	0.00	0.08	1.46	1.72	3.26	3.32	1.70	0.12	2.16
	EMGNN	7.08	7.10	4.22	3.30	1.46	0.84	3.74	3.92	5.00	5.12	4.22	**1.36**	3.95

2.3 Individual-PPI network performance

To evaluate the impact of integrating multiple protein-protein interaction (PPI) networks, the full multi-network NAMAT model was compared with variants trained on each individual PPI network. Supplementary Table S4 presents the area under the precision-recall curve (AUPRC) for NAMAT and each single-PPI model across 12 cancer types. NAMAT consistently outperformed all individual-PPI models for every cancer type. While PCNet yielded the highest average performance among the single-PPI models, NAMAT further increased the average AUPRC from 0.635 to 0.683. These findings demonstrate that the improvement achieved by NAMAT is attributable to adaptive multi-network integration rather than reliance on a single strong PPI network.

Table S4. AUPRC comparison between NAMAT and models trained on individual PPI networks across 12 cancer types. Results are reported as mean AUPRC ± standard deviation over repeated runs, rounded to three decimals. IRef and IRef2015 denote IRefIndex and IRefIndex_2015, respectively. The best result for each cancer type is shown in **bold**.

Cancer	NAMAT	CPDB	STRING	MultiNet	PCNet	IRef	IRef2015
BRCA	**0.802**±0.006	0.705±0.008	0.704±0.006	0.720±0.006	0.750±0.009	0.686±0.008	0.717±0.009
BLCA	**0.789**±0.005	0.743±0.010	0.727±0.008	0.727±0.007	0.750±0.005	0.705±0.010	0.740±0.008
LUAD	**0.744**±0.008	0.681±0.008	0.651±0.008	0.634±0.013	0.706±0.007	0.664±0.006	0.663±0.008
LIHC	**0.595**±0.019	0.538±0.009	0.532±0.015	0.524±0.017	0.576±0.017	0.488±0.009	0.509±0.015
THCA	**0.426**±0.026	0.341±0.015	0.299±0.016	0.305±0.015	0.351±0.024	0.291±0.024	0.345±0.017
LUSC	**0.616**±0.053	0.413±0.020	0.420±0.041	0.411±0.047	0.539±0.031	0.475±0.040	0.361±0.048
ESCA	**0.611**±0.015	0.535±0.016	0.522±0.010	0.498±0.015	0.557±0.018	0.535±0.011	0.530±0.011
PRAD	**0.827**±0.009	0.723±0.019	0.755±0.014	0.764±0.035	0.744±0.009	0.742±0.020	0.754±0.017
STAD	**0.779**±0.009	0.718±0.012	0.713±0.006	0.700±0.012	0.759±0.010	0.728±0.014	0.723±0.011
COAD	**0.610**±0.008	0.575±0.007	0.566±0.010	0.553±0.007	0.588±0.005	0.567±0.009	0.570±0.009
UCEC	**0.743**±0.014	0.710±0.015	0.707±0.011	0.723±0.011	0.736±0.012	0.726±0.015	0.733±0.017
CESC	**0.655**±0.030	0.534±0.052	0.541±0.042	0.577±0.050	0.565±0.043	0.505±0.036	0.599±0.021

Incremental contribution of individual PPI networks

To further investigate the contribution of individual protein–protein interaction (PPI) networks, a greedy incremental network selection analysis was conducted on the BLCA dataset. Initially, the network with the highest AUPRC in Supplementary Table S4, PCNet, was selected as the starting configuration. At each subsequent step, each remaining network was individually added to the current best-performing combination, and the network resulting in the largest improvement in AUPRC was retained. This process continued until all six PPI networks had been evaluated.

Supplementary Table S5 demonstrates that integrating additional protein-protein interaction (PPI) networks progressively enhanced performance during the initial stages of network fusion. Beginning with PCNet alone (AUPRC = 0.7500), the inclusion of MultiNet resulted in the largest performance gain among all candidate networks (+0.0177, $p = 7.02 \times 10^{-8}$), suggesting that MultiNet contributes complementary information beyond that provided by PCNet. Using the same greedy selection approach, CPDB and STRING were subsequently added, leading to an AUPRC of 0.7893 when PCNet, MultiNet, CPDB, and STRING were combined.

In contrast, after the integration of these four networks, the addition of IRefIndex or IRefIndex2015 resulted in only marginal changes. Incorporating IRefIndex into the PCNet+MultiNet+CPDB+STRING configuration produced a negligible change in AUPRC (-0.0002, $p = 0.3616$), while the inclusion of IRefIndex2015 had a similarly minimal effect (-0.0017, $p = 0.2188$). Expanding the model from five to all six networks did not yield a statistically significant improvement ($p = 0.2680$).

These results indicate that NAMAT benefits from the integration of complementary information across multiple PPI networks, especially during the initial stages of network fusion. The observed saturation effect suggests that the majority of relevant information for BLCA is captured by the combination of PCNet, MultiNet, CPDB, and STRING, while additional networks offer limited incremental benefit. Overall, this analysis supports the effectiveness of NAMAT's multi-network integration strategy and highlights diminishing returns as more redundant interaction networks are incorporated.

2.4 Detailed ablation results

Supplementary Table S6 presents the complete per-cancer ablation results. Each block compares two matched configurations that differ only in the component under investigation, enabling a controlled assessment of the contributions from message-aware attention, input feature transformation, residual connections, and graph-based message passing. The full NAMAT model

Table S5. Incremental contribution of individual PPI networks to NAMAT performance on BLCA. Network indices denote individual PPI networks: 1 = CPDB, 2 = STRING, 3 = MultiNet, 4 = PCNet, 5 = IRefIndex, and 6 = IRefIndex2015.

Baseline configuration	Added network	AUPRC after	Mean diff.	*p*-value
4 (0.7500±0.0054)	+1 (CPDB)	0.7614±0.0071	+0.0112	0.0007
	+2 (STRING)	0.7658±0.0069	+0.0156	5.82×10^{-6}
	+3 (MultiNet)	0.7679±0.0070	+0.0177	7.02×10^{-8}
	+5 (IRefIndex)	0.7604±0.0077	+0.0102	4.96×10^{-5}
	+6 (IRefIndex2015)	0.7606±0.0075	+0.0104	4.73×10^{-5}
4+3 (0.7679±0.0070)	+1 (CPDB)	0.7818±0.0072	+0.0139	8.45×10^{-6}
	+2 (STRING)	0.7791±0.0071	+0.0112	2.39×10^{-6}
	+5 (IRefIndex)	0.7798±0.0055	+0.0119	1.26×10^{-6}
	+6 (IRefIndex2015)	0.7808±0.0051	+0.0129	4.44×10^{-6}
4+3+1 (0.7818±0.0072)	+2 (STRING)	0.7893±0.0066	+0.0075	0.0001
	+5 (IRefIndex)	0.7835±0.0057	+0.0017	0.0495
	+6 (IRefIndex2015)	0.7839±0.0059	+0.0021	0.0242
4+3+1+2 (0.7893±0.0066)	+5 (IRefIndex)	0.7891±0.0070	-0.0002	0.3616
	+6 (IRefIndex2015)	0.7876±0.0055	-0.0017	0.2188
4+3+1+2+5 (0.7891±0.0070)	+6 (IRefIndex2015)	0.7890±0.0050	-0.0001	0.2680

achieved the highest average AUPRC. Excluding graph-based message passing resulted in the most significant performance decline, with average AUPRC decreasing from 0.683 to 0.516. Incorporating residual connections yielded a substantial improvement, increasing average AUPRC from 0.590 to 0.683. The input MLP consistently enhanced performance, and message-aware attention provided further gains compared to uniform graph weighting.

2.5 Biological relevance analysis

To determine whether the top-ranked NAMAT predictions represent biologically meaningful signals, three complementary evaluation strategies were employed: over-representation analysis (ORA), pathway-level consistency analysis using gene set enrichment analysis (GSEA), and validation against independent disease-evidence databases.

Functional enrichment analysis. ORA was first conducted on the top-ranked genes for CESC, LUSC, and PRAD using Human Disease Ontology (HDO) and KEGG gene sets. This analysis evaluates whether disease-relevant gene sets are statistically over-represented among the top-ranked predictions.

Supplementary Table S7 demonstrates that NAMAT predictions are enriched for disease-relevant Human Disease Ontology (HDO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) terms. Fold enrichment values ranged from 2.99 to 10.04, with the majority of results exhibiting strong statistical significance. Prostate adenocarcinoma (PRAD) displayed consistent enrichment for both prostate cancer HDO and KEGG terms, whereas lung squamous cell carcinoma (LUSC) exhibited particularly strong KEGG enrichment. In cervical squamous cell carcinoma and endocervical adenocarcinoma (CESC), enrichment of the HPV-related KEGG pathway hsa05165 aligns with the established biological role of HPV-associated processes in cervical cancer.

Pathway-level consistency analysis. The full ranked gene list was subsequently evaluated using Gene Set Enrichment Analysis (GSEA). In contrast to Over-Representation Analysis (ORA), which considers only a fixed top-K subset, GSEA assesses whether genes from predefined disease or pathway gene sets are concentrated near the top of the complete ranked list.

Supplementary Table S8 demonstrates that disease-associated gene sets are strongly enriched near the top of the NAMAT-ranked gene lists. All evaluated terms yielded positive normalized enrichment score (NES) values, with NES reaching up to 4.29 and highly significant false discovery rate (FDR) values. The leading-edge fractions indicate that disease-relevant genes contribute substantially to the enrichment signals, thereby supporting the ranking-level biological consistency of NAMAT predictions.

Validation using independent databases. Top-ranked predictions were cross-checked against independent disease-gene association databases. This analysis assessed whether genes prioritized by NAMAT overlap with known disease-associated genes from resources not used as model labels.

Table S6. Comprehensive ablation study across 12 cancer types. Each block presents a controlled comparison isolating the effect of a specific component. `MAA`: message-aware attention; `MLP`: input feature transformation; `Res`: residual connection; `MSG`: graph-specific message information. Results are reported as mean AUPRC ± standard deviation over repeated runs, rounded to three decimals.

Panel A. BRCA, BLCA, LUAD, LIHC, THCA, and LUSC.

Test	Configuration	BRCA	BLCA	LUAD	LIHC	THCA	LUSC
MAA	−MAA −MLP −Res	0.670±0.006	0.616±0.008	0.499±0.012	0.462±0.023	0.315±0.014	**0.385**±0.041
	+MAA −MLP −Res	**0.736**±0.008	**0.660**±0.013	**0.559**±0.017	**0.498**±0.027	**0.327**±0.015	0.365±0.036
MAA	−MAA +MLP −Res	0.716±0.006	0.667±0.006	0.583±0.011	0.505±0.018	0.359±0.017	**0.490**±0.023
	+MAA +MLP −Res	**0.750**±0.005	**0.692**±0.013	**0.615**±0.011	**0.515**±0.014	**0.364**±0.018	0.462±0.031
MLP	−MAA −MLP −Res	0.670±0.006	0.616±0.008	0.499±0.012	0.462±0.023	0.315±0.014	0.385±0.041
	−MAA +MLP −Res	**0.716**±0.006	**0.667**±0.006	**0.583**±0.011	**0.505**±0.018	**0.359**±0.017	**0.490**±0.023
MLP	+MAA −MLP −Res	0.736±0.008	0.660±0.013	0.559±0.017	0.498±0.027	0.327±0.015	0.365±0.036
	+MAA +MLP −Res	**0.750**±0.005	**0.692**±0.013	**0.615**±0.011	**0.515**±0.014	**0.364**±0.018	**0.462**±0.031
Res	+MAA +MLP −Res	0.750±0.005	0.692±0.013	0.615±0.011	0.515±0.014	0.364±0.018	0.462±0.031
	+MAA +MLP +Res	**0.802**±0.006	**0.789**±0.005	**0.744**±0.008	**0.595**±0.019	**0.426**±0.026	**0.616**±0.053
MSG	−MSG +MLP +Res	0.513±0.004	0.653±0.009	0.582±0.006	0.440±0.010	0.233±0.014	0.412±0.025
	+MSG +MLP +Res	**0.802**±0.006	**0.789**±0.005	**0.744**±0.008	**0.595**±0.019	**0.426**±0.026	**0.616**±0.053

Panel B. ESCA, PRAD, STAD, COAD, UCEC, and CESC.

Test	Configuration	ESCA	PRAD	STAD	COAD	UCEC	CESC
MAA	−MAA −MLP −Res	0.403±0.012	0.639±0.013	0.570±0.015	0.449±0.014	**0.523**±0.014	0.498±0.039
	+MAA −MLP −Res	**0.448**±0.023	**0.665**±0.021	**0.624**±0.009	**0.456**±0.019	0.522±0.010	**0.587**±0.062
MAA	−MAA +MLP −Res	0.479±0.014	0.726±0.019	0.609±0.013	0.497±0.007	0.560±0.014	0.643±0.047
	+MAA +MLP −Res	**0.506**±0.019	**0.747**±0.022	**0.646**±0.010	**0.508**±0.008	**0.596**±0.015	**0.677**±0.050
MLP	−MAA −MLP −Res	0.403±0.012	0.639±0.013	0.570±0.015	0.449±0.014	0.523±0.014	0.498±0.039
	−MAA +MLP −Res	**0.479**±0.014	**0.726**±0.019	**0.609**±0.013	**0.497**±0.007	**0.560**±0.014	**0.643**±0.047
MLP	+MAA −MLP −Res	0.448±0.023	0.665±0.021	0.624±0.009	0.456±0.019	0.522±0.010	0.587±0.062
	+MAA +MLP −Res	**0.506**±0.019	**0.747**±0.022	**0.646**±0.010	**0.508**±0.008	**0.596**±0.015	**0.677**±0.050
Res	+MAA +MLP −Res	0.506±0.019	0.747±0.022	0.646±0.010	0.508±0.008	0.596±0.015	**0.677**±0.050
	+MAA +MLP +Res	**0.611**±0.015	**0.827**±0.009	**0.779**±0.009	**0.610**±0.008	**0.743**±0.014	0.655±0.030
MSG	−MSG +MLP +Res	0.456±0.011	0.579±0.014	0.676±0.006	0.526±0.006	0.678±0.009	0.447±0.032
	+MSG +MLP +Res	**0.611**±0.015	**0.827**±0.009	**0.779**±0.009	**0.610**±0.008	**0.743**±0.014	**0.655**±0.030

Table S7. ORA of top-K predicted genes (K= 100–200) using HDO and KEGG. Count denotes the number of overlapping genes, FE denotes fold enrichment, and FDR was computed using the Benjamini-Hochberg correction.

Cancer	Term	Collection	K	Count	FE	FDR (BH)
CESC	Cervical cancer	HDO	100	9	7.03	3.87×10^{-5}
			150	13	6.63	7.39×10^{-7}
			200	16	6.25	4.86×10^{-8}
	hsa05165	KEGG	100	15	3.48	1.18×10^{-4}
			150	19	2.99	8.77×10^{-5}
			200	27	3.23	3.64×10^{-7}
LUSC	Lung SCC	HDO	100	3	4.53	9.01×10^{-2}
			150	3	2.99	1.66×10^{-1}
			200	5	3.75	3.97×10^{-2}
	hsa05223	KEGG	100	10	10.04	7.17×10^{-7}
			150	12	8.17	2.67×10^{-7}
			200	15	7.77	6.63×10^{-9}
PRAD	Prostate cancer	HDO	100	20	4.79	3.67×10^{-7}
			150	29	4.63	2.52×10^{-10}
			200	37	4.43	4.56×10^{-13}
	hsa05215	KEGG	100	10	8.91	2.35×10^{-6}
			150	14	8.16	1.86×10^{-8}
			200	19	8.41	1.05×10^{-11}

Table S8. Gene Set Enrichment Analysis (GSEA) of the ranked gene list using Human Disease Ontology (HDO) and Kyoto Encyclopedia of Genes and Genomes (KEGG). NES denotes normalized enrichment score, FDR was computed using the Benjamini-Hochberg correction, and LE denotes leading-edge fractions.

Cancer	Term/Pathway	Collection	Set size	NES	FDR (BH)	LE (%)
CESC	Cervical SCC	HDO	21	2.65	1.66×10^{-4}	38/6
	hsa05165	KEGG	290	4.29	3.54×10^{-48}	48/13
LUSC	Lung SCC	HDO	46	2.22	2.51×10^{-5}	35/12
	hsa05223	KEGG	65	3.49	1.91×10^{-21}	80/20
PRAD	Prostate cancer	HDO	286	3.54	4.71×10^{-66}	61/16
	hsa05215	KEGG	84	3.68	4.02×10^{-33}	74/14

Table S9. DisGeNET validation of top-K predicted genes (K=50). Count denotes the number of overlapping genes, P@K denotes precision at K, FE denotes fold enrichment, and FDR was computed using Benjamini–Hochberg correction.

Cancer	K	Count	P@K	FE	FDR (BH)
CESC	50	2	0.04	16.32	1.32×10^{-2}
LUSC	50	3	0.06	13.43	1.89×10^{-3}
PRAD	50	9	0.18	4.91	9.56×10^{-5}

Table S10. Open Targets validation of top-K predicted genes (K=50, score ≥ 0.20). Count denotes the number of overlapping genes, P@K denotes precision at K, FE denotes fold enrichment, and FDR was computed using Benjamini–Hochberg correction.

Cancer	Threshold	K	Count	P@K	FE	FDR (BH)
CESC	0.20	50	25	0.50	6.78	1.02×10^{-15}
LUSC	0.20	50	16	0.32	6.45	1.48×10^{-9}
PRAD	0.20	50	13	0.26	5.70	3.17×10^{-7}

Supplementary Tables S9 and S10 demonstrate that the top-ranked NAMAT predictions overlap significantly with established disease-associated genes. In DisGeNET, CESC and LUSC exhibit high fold enrichment, although the absolute overlap counts remain low, likely due to limited disease annotations. In contrast, Open Targets displays higher overlap counts, with 25 CESC-associated genes, 16 LUSC-associated genes, and 13 PRAD-associated genes identified among the top 50 predictions at a score threshold of ≥ 0.20.

Overall, the concordance among functional enrichment, pathway-level consistency, and independent disease-evidence databases suggests that NAMAT predictions are predictive with respect to benchmark labels and are also biologically meaningful. These findings indicate that message-aware multi-graph integration captures disease-relevant biological signals, rather than generating predictions solely based on benchmark-specific label distributions.

2.6 Interpretation of node-wise graph reliability in NAMAT

To elucidate the prediction mechanism of NAMAT, this study investigates how the model integrates information from multiple PPI networks during message passing. Three complementary aspects of the learned model behavior are analyzed: (i) gene-specific graph reliability patterns, which indicate how different genes assign varying reliability scores to distinct interaction networks; (ii) network contribution scores, which quantify the magnitude of information propagated from each network following attention weighting; and (iii) neighbor-level evidence, which identifies influential neighboring genes that support each prediction. Collectively, these analyses offer an interpretable perspective on how NAMAT utilizes heterogeneous biological networks to prioritize cancer driver genes.

2.6.1 Gene-specific graph reliability patterns

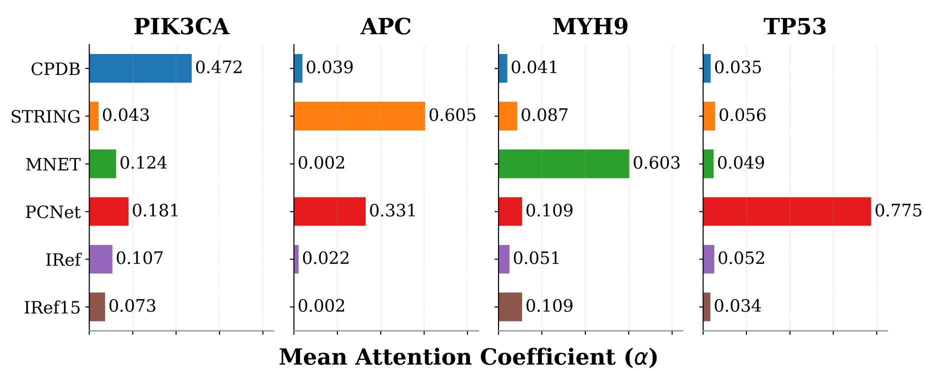

Figure S1. Graph reliability scores learned by NAMAT for representative driver genes in stomach adenocarcinoma (STAD). Bars represent mean attention coefficients averaged across 10 repeated runs.

The analysis first assessed whether NAMAT learns gene-specific preferences across interaction networks by examining the graph reliability scores assigned to representative driver genes in stomach adenocarcinoma (STAD). Figure S1 presents the mean attention coefficients learned for PIK3CA, APC, MYH9, and TP53. The attention distributions differ substantially across genes, suggesting that NAMAT does not apply a fixed global weighting to PPI networks. Rather, each gene exhibits a distinct reliability profile across the available interaction sources. These examples support the central design motivation of NAMAT: the relevance of a PPI network is context-dependent and should be learned at the node level, not shared globally across all genes.

2.6.2 Network contribution analysis

The subsequent analysis examined how information from different interaction networks contributes to node representation learning. For each gene and PPI network, the network contribution was summarized as the magnitude of the graph-specific

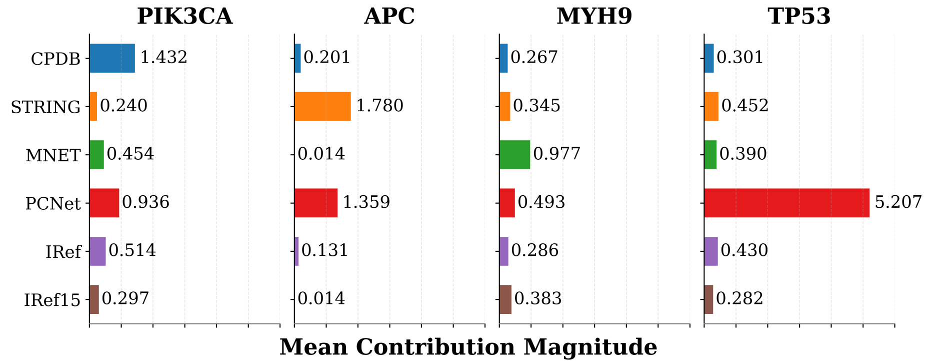

Figure S2. Network contribution scores learned by NAMAT for representative driver genes in stomach adenocarcinoma (STAD). Bars represent mean contribution magnitudes averaged across 10 repeated runs.

message after attention weighting. Consequently, the contribution score reflects both the reliability assigned to a network and the extent to which it provides a substantial propagated message for a given gene.

Figure S2 presents the mean contribution magnitudes associated with PIK3CA, APC, MYH9, and TP53. Consistent with the graph reliability patterns, the contribution distributions vary substantially across genes, indicating that different genes receive information from distinct combinations of PPI networks. Furthermore, the overall contribution patterns align with the graph reliability distributions observed in Figure S1, suggesting that networks assigned higher reliability typically contribute more strongly during message aggregation. These findings further support the importance of node-wise graph reliability estimation in adaptive multi-graph integration.

2.6.3 Neighbor-level evidence supporting driver gene prediction

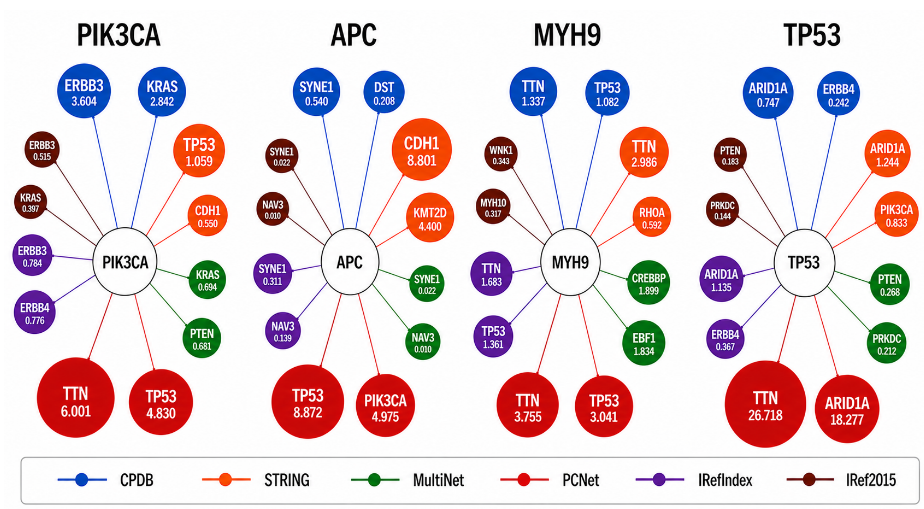

Figure S3. Top contributing neighbors identified by NAMAT for representative driver genes in stomach adenocarcinoma (STAD). Node size is proportional to the mean contribution score, averaged across 10 repeated runs.

To further investigate the evidence utilized by NAMAT during prediction, the most influential neighboring genes contributing to the representations of representative driver genes were analyzed. Figure S3 presents the top contributing neighbors identified from the six PPI networks, with node size proportional to the corresponding contribution score.

Several meaningful patterns emerge from the analysis. For instance, ERBB3, KRAS, and TP53 are among the top contributing neighbors for PIK3CA, while TP53, PIK3CA, and CDH1 provide strong neighbor-level evidence for APC. Additionally, recurrent neighbors such as TTN, TP53, ARID1A, and PTEN are identified for MYH9 and TP53 across multiple interaction networks.

These findings indicate that NAMAT is capable of revealing local network evidence supporting its predictions. It is important to interpret the identified neighbors as model-derived supporting evidence rather than as direct causal mechanisms. The recurrence of known cancer-associated genes among the top contributing neighbors further demonstrates that NAMAT can extract biologically relevant signals from heterogeneous PPI sources. Collectively, this analysis demonstrates that NAMAT provides interpretable information at multiple levels, including graph-level reliability, network-level contribution, and neighbor-level evidence.

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